A set $F[x] = \{ \sum_{i=0}^{n} a_i \cdot x^i \mid n \in \mathbb{N},\ a_i \in F \}$ with addition and scalar-multiplication is called the polynomial space. As we know, it is countably infinite dimensional,

$$F[x] = \mathrm{Span}_F \{ x^k \mid k \in \mathbb{N} \cup \{0\} \}.$$

In this paper, we investigate some typical properties of the polynomial space, using our background knowledge about dual, normal form, and inner product.

In $\mathbb{C}[x]$, the map

$$\langle \cdot , \cdot \rangle : \mathbb{C}[x] \times \mathbb{C}[x] \to \mathbb{C} : \int_I f(x) \cdot \overline{g(x)}\, dx$$ where I is closed interval

is an inner Product on $\mathbb{C}[x]$, because: given $f, h, g \in \mathbb{C}[x]$, and $c \in \mathbb{C}$,

$$\int_I [cf + h] \cdot \overline{g} = c \int_I f \cdot \overline{g} + \int_I h \cdot \overline{g}$$ by the linearity of integral

And

$$\overline{\int_I f \cdot \overline{g}} = \int_I \overline{f \cdot \overline{g}} = \int_I \overline{f} \cdot g = \int_I g \cdot \overline{f},$$ because

$$\overline{\int_I f} = \overline{\int_I \mathrm{Re} f + i \cdot \int_I \mathrm{Im} f} = \int_I \mathrm{Re} f - i \cdot \int_I \mathrm{Im} f = \int_I \mathrm{Re} f - i \cdot \mathrm{Im} f = \int_I \overline{f}.$$

Finally,

$$\int_I f \cdot \overline{f} = \int_I \|f\|^2 \geq 0.$$ So, shown. ☒

「Def. Let $V = \mathbb{C}[x]$, the space of complex coefficient polynomials.
Define an inner product on V as;

$$\langle f, g \rangle = \int_0^1 f(x)\overline{g(x)}\,dx,$$ where $\overline{g}$ is obtained by take conjugate to coeff. g.

1. V does not satisfy the Riesz-representation Theorem.

Proof. Let $z \in \mathbb{C}$ be fixed. Let $L : V \to \mathbb{C} : f \mapsto f(z)$ be the evaluation at z.
Clearly, L is a linear functional. Suppose that for some $g \in V$ s.t

$$\langle f, g \rangle = L(f) = f(z) \text{ for all } f \in V.$$

Take $h(x) = x - z$. Then, for any $f \in V$,

$$\langle h \cdot f, g \rangle = \int_0^1 h(t) \cdot f(t) \cdot \overline{g(t)}\,dt = h(z) \cdot g(z) = 0,$$

Put $f = \overline{h}g$, then

$$\int_0^1 |h(t)|^2 |g(t)|^2 dt = 0,$$ so $h(t)g(t) = 0$, but $h(t) \neq 0$, so $g \equiv 0$.

This impies $L \equiv 0$, but it is contradiction.

2. Adjoint of a linear operator on V may not exist.

「Finite subspace of $F[x]$.

Let $P_2(\mathbb{R}) = \text{Span}_{\mathbb{R}}\{1, x, x^2\}$, equipped inner product

$$\langle f, g\rangle = \int_0^1 f(t)\cdot g(t)\, dt.$$

1). For fixed $a \in \mathbb{R}$, let $V_a : P_2(\mathbb{R}) \to \mathbb{R} : f \mapsto f(a)$.
Find a polynomial $h \in P_2(\mathbb{R})$ s.t.

$$\langle f, h\rangle = V_a(f) = f(a) \quad \text{for all } f \in P_2(\mathbb{R}).$$

Proof. Construct an orthonormal basis for $P_2(\mathbb{R})$ as:

$\alpha_1 = 1.$

$$\alpha_2 = x - \frac{\langle x, 1\rangle}{\|1\|^2}\cdot 1 = x - \frac{1}{2},$$

$$\alpha_3 = x^2 - \frac{\langle x^2, 1\rangle}{\|1\|^2}\cdot 1 - \frac{\langle x^2, x - \frac{1}{2}\rangle}{\|x - \frac{1}{2}\|^2}\cdot\left(x - \frac{1}{2}\right) = x^2 - \frac{1}{3} - \cdot\left(x - \frac{1}{2}\right)$$

$$= x^2 - x + \frac{1}{6}.$$

$$\langle x^2, x - \tfrac{1}{2}\rangle = \int_0^1 x^3 - \frac{1}{2}x^2\, dx = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}.$$

$$\langle x - \tfrac{1}{2}, x - \tfrac{1}{2}\rangle = \int_0^1 \left(x - \tfrac{1}{2}\right)^2 dx = 2\cdot\int_0^{\frac{1}{2}} x^2\, dx = 2\cdot\frac{1}{3}\cdot\frac{1}{8} = \frac{1}{12}$$

$$\langle x^2 - x + \tfrac{1}{6}, x^2 - x + \tfrac{1}{6}\rangle = \int_0^1 x^4 - x^3 + \frac{1}{6}x^2 - x^3 + x^2 - \frac{1}{6}x + \frac{1}{6}x^2 - \frac{1}{6}x + \frac{1}{36}\, dx$$

$$= \int_0^1 x^4 - 2x^3 + \frac{4}{3}x^2 - \frac{1}{3}x + \frac{1}{36}\, dx$$

$$= \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} = \frac{1}{180}.$$

Therefore, $\left\{1, \sqrt{12}\left(x - \tfrac{1}{2}\right), \sqrt{180}\cdot\left(x^2 - x + \tfrac{1}{6}\right)\right\}$ is an orthonormal basis.

Now, $$h(x) = \overline{1}\cdot 1 + \overline{\sqrt{12}\cdot\left(a - \tfrac{1}{2}\right)}\cdot\sqrt{12}\left(x - \tfrac{1}{12}\right) + \overline{\sqrt{180}\cdot\left(a^2 - a + \tfrac{1}{6}\right)}\cdot\sqrt{180}\left(x^2 - x + \tfrac{1}{6}\right).$$

$$= 1 + (12a - 6)\left(x - \tfrac{1}{12}\right) + 180\left(a^2 - a + \tfrac{1}{6}\right)\left(x^2 - x + \tfrac{1}{6}\right)$$

is that desired.